

A Guide to Statistical Network Modeling

Session 3.1: SAOMs for Actor-Driven Network Dynamics

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instats Seminar

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Companion computations and reproducibility

- **Shared resources:** [repository](#) and [companion app](#); see Session 1.1 for the common reproducibility workflow.
- Open **Session 3.1** and choose one of **five visible public network panels**; the page exposes the support rule, model boundary, estimates, convergence diagnostics, simulation envelopes, and a downloadable computation record.
- **Computation boundary:** the app runs a stated RSiena network-only actor-oriented model; it does not represent a fitted coefficient as a causal effect or silently fill structural absence with a non-tie.

Learning objectives

By the end of this session, we will be able to:

- define a **stochastic actor-oriented model (SAOM)** for a repeated binary network;
- distinguish the **rate function**, the **evaluation function**, and the actor's discrete choice set;
- interpret density, reciprocity, and closure effects at the level of a simulated actor microstep;
- state the method-of-moments target underlying RSiena estimation; and
- audit a fitted network-only SAOM through convergence diagnostics and simulation-based goodness of fit.

Session route

- ① **Data and process:** repeated observations, actor support, and continuous-time microsteps.
- ② **Model construction:** rates, objective functions, and structural network effects.
- ③ **Estimation and diagnostics:** moments, convergence, and simulated network checks.
- ④ **No-code practice:** five public panels and a support-aware upload workflow.

From observations to an actor-oriented process

- We observe network states at discrete occasions $t_1 < \dots < t_M$, but posit unobserved change between them.
- A **stochastic actor-oriented model** is a continuous-time Markov model in which actors receive opportunities to alter their own outgoing ties.
- The data do not reveal the actual order of microsteps; that order is **simulated** under the fitted process.
- The resulting model is appropriate for a repeated network with a defensible actor set, tie meaning, observation scale, and support.

The network state

- Let $X(t) = (X_{ij}(t))$ denote a binary network at process time t on actor set $\mathcal{N}$.
- For a directed network, $X_{ij}(t) = 1$ means that actor i sends the documented tie to actor j ; $X_{ij}(t) = 0$ means no such tie is recorded.
- Self-ties are excluded: $X_{ii}(t) = 0$.
- In an undirected network, $X_{ij}(t) = X_{ji}(t)$; the admissible microstep mechanism must preserve this symmetry.
- **Critical boundary:** the value zero is a documented non-tie, not an unobserved or structurally unavailable dyad.

Observation waves are not microsteps

- A **wave** is an observed network snapshot, such as a survey occasion or annual treaty inventory.
- The interval (t_m, t_{m+1}) contains the modeled, unobserved actor opportunities and tie changes.
- Two waves identify an interval but provide limited leverage for distinguishing stable effects from transient mechanisms.
- More waves improve the description of change, but do not by themselves solve omitted-variable, measurement, or population-boundary problems.

Actor support and the target population

- The **actor support** specifies which actors and dyads may be represented at each observation occasion.
- A balanced teaching panel retains actors observed at every selected wave; this is an explicit restriction, not a claim that excluded actors are isolates.
- A composition-change model can represent entry and exit, but requires documented entry times and RSiena composition information.
- The app's public workflows state whether they use an all-wave balanced panel and display the support rule before fitting.

Tie response: binary and substantively defined

- The Session 3.1 calculation uses a **binary tie response**: each admissible dyad is present or absent at a wave.
- The tie must have a common interpretation across selected waves: for example, friendship nomination, advice, or alliance presence.
- Collapsing a valued or ordinal relation to binary discards information and must be stated as a modeling choice.
- Structural missingness, nonresponse, and actor absence require explicit handling; they must not be converted to zeros for convenience.

Rate functions: who receives an opportunity?

- Actor i receives network-change opportunities at rate $\lambda_i^X(x; \rho) > 0$ when the current state is x .
- A larger rate implies more expected opportunities for that actor or period; it is not a tie coefficient and is not an observed event count.
- A standard RSiena fit estimates **period-specific rate parameters** because observation intervals can differ in overall activity.
- Conditional on receiving an opportunity, the actor evaluates the permitted alternatives through the objective function.

An actor microstep

- First, a rate mechanism selects an actor for a network microstep.
- Second, the selected actor compares the current outgoing-tie configuration with permitted one-tie alternatives.
- A microstep may leave the state unchanged; a realized opportunity does not force a tie toggle.
- The model therefore separates **how often opportunities occur** from **which local network state is preferred**.

The actor's choice set

- Write $\mathcal{A}_i(x)$ for states available to actor i from current state x .
- For a directed binary network, $\mathcal{A}_i(x)$ contains x and states that differ from x by toggling one outgoing tie X_{ij} .
- The choice set respects documented constraints: no loops, declared dyad support, and any imposed structural zeros.
- It does not permit arbitrary simultaneous rewiring of several ties by one actor at one microstep.

Choice probabilities from an objective function

- For $x' \in \mathcal{A}_i(x)$, the standard multinomial-logit choice rule is

$$\mathbb{P}\{X(t^+) = x' \mid i, x\} = \frac{\exp\{f_i^X(x'; \beta)\}}{\sum_{z \in \mathcal{A}_i(x)} \exp\{f_i^X(z; \beta)\}}.$$

- t^+ denotes the state immediately after the selected actor's opportunity.
- This multinomial-logit form corresponds to the standard random-utility formulation in which stochastic disturbances accompany the systematic evaluation function.
- The normalization is over **the actor's local alternatives**, not over independent dyads in the whole network.
- This equation defines a stochastic process; it is not an ordinary cross-sectional logistic regression.

The evaluation function

- The **network evaluation function** is

$$f_i^X(x; \beta) = \sum_{k=1}^K \beta_k s_{ik}^X(x),$$

where $s_{ik}^X(x)$ is actor i 's contribution to network statistic k .

- A positive β_k makes alternatives with larger s_{ik}^X relatively more attractive, conditional on the other terms and the choice set.
- The statistical meaning is local and conditional: it concerns relative microstep choices, not a marginal population risk ratio.

Change statistics explain a local effect

- For a potential toggle $x \rightarrow x'$, define the **change statistic** $\Delta_{ik}s_k = s_{ik}^X(x') - s_{ik}^X(x)$.
- The change in evaluation is $f_i^X(x'; \beta) - f_i^X(x; \beta) = \sum_k \beta_k \Delta_{ik}s_k$.
- Thus an effect contributes through the structural change produced by a specific tie decision.
- A coefficient's sign alone is insufficient without the response definition, competing effects, support, and relevant change statistic.

Density effect

- The **density effect** counts outgoing ties for directed networks, or incident ties for undirected networks.
- It provides a baseline tendency toward sparse or dense states after the rate process has selected an opportunity.
- A negative density parameter is common in sparse networks, but its numerical magnitude depends on the remaining effects and the microstep choice scale.
- Density is not a descriptive density percentage and must not be interpreted as one.

Reciprocity in directed networks

- **Reciprocity** rewards a choice $i \rightarrow j$ when the reverse tie $j \rightarrow i$ is already present.
- The statistic is defined only for directed relations with a meaningful reverse dyad.
- A positive reciprocity parameter favors mutual local configurations relative to the model's other alternatives.
- It does not imply that each recipient consciously reciprocates, nor does it establish a causal response to an observed tie.

Transitive triplets in directed networks

- A **transitive triplet** is supported when $i \rightarrow j$, $j \rightarrow h$, and $i \rightarrow h$ form a directed closure pattern.
- The effect rewards an actor's tie that completes locally transitive directed structure.
- It differs from simple reciprocity: a closure path may involve three distinct actors and need not be mutual.
- Strong closure terms can interact with density and other effects; convergence and simulated triad diagnostics are essential.

Directed cycles are a different statistic

- A **directed cycle** has orientation $i \rightarrow j$, $j \rightarrow h$, and $h \rightarrow i$.
- A directed transitive triplet has orientation $i \rightarrow j$, $j \rightarrow h$, and $i \rightarrow h$.
- The two patterns encode different mechanisms and cannot be interchanged because both involve three nodes.
- A directed triad census in goodness-of-fit checks whether the fitted model reproduces the observed mixture of such patterns.

Transitive triads in undirected networks

- In an undirected network, a **transitive triad** is a closed triangle: all three pairwise ties are present.
- There is no direction-specific reciprocity term because $X_{ij} = X_{ji}$ by construction.
- The COW public workflow therefore uses density and the RSiena **transitive-triads** closure effect, not a directed-triplet effect.
- An undirected annual network still requires a clear actor support and observation-interval interpretation.

What is identified by repeated observations?

- Identification concerns whether the stated parameter vector can be distinguished from alternatives using the modeled repeated-network process.
- Sparse data, few waves, near-deterministic stability, or nearly collinear effects can yield weak or unstable information.
- A two-wave panel gives one transition interval; it can support a limited dynamic specification but offers little leverage for complex time variation.
- Formal estimation output cannot repair an ill-defined population, ambiguous tie meaning, or unsupported structural effect.

Observed statistics become moment targets

- Let $m(\cdot)$ be a moment vector summarizing changes across the observed periods, such as tie count, mutuality, and closure contributions.
- **Method of moments** seeks $\hat{\theta}$ such that

$$\mathbb{E}_{\hat{\theta}}\{m(X_{\text{sim}}) \mid X(t_1) = x(t_1)\} \approx m(x_{\text{obs}}).$$

- The expectation is computed by simulations from the actor-oriented model, not by assuming dyads are independent.
- The selected effects determine which model statistics are directly targeted; omitted statistics remain available for goodness-of-fit checks.

Stochastic approximation in RSiena

- RSiena iteratively simulates the proposed process, compares simulated and observed moments, and updates parameter values.
- The update is stochastic because the simulated moments vary across runs.
- A **pilot fit** with small simulation settings is useful for teaching and debugging, but is not a substitute for a documented final estimation run.
- The app exposes its random seed, RSiena n_3 setting, and number of goodness-of-fit simulations in the downloadable record.

Convergence is an audit, not a decorative number

- RSiena reports convergence t -ratios for individual effects and an overall maximum convergence ratio after estimation.
- A common practical reference is maximum absolute ratio below 0.25; a smaller value, often below 0.10, is preferable for a final analysis.
- These are diagnostic guidelines, not proofs that the model is correctly specified.
- Nonconvergence, warning messages, or implausible estimates require model/data review rather than selective reporting.

A SAOM parameter is not a causal effect

- A fitted structural parameter summarizes the stated stochastic choice model conditional on included effects and data support.
- It does not automatically identify an intervention effect, individual motive, or causal pathway.
- Causal claims require a design and assumptions that address confounding, timing, interference, measurement, and population definition.
- The app labels its computations as model-based associations and retains source-specific interpretation limits beside each workflow.

Simulating fitted network states

- After fitting, RSiena can simulate repeated network outcomes from the estimated actor-oriented process.
- We compare observed statistics with the distribution of corresponding statistics in these simulated networks.
- A close match on fitted statistics is expected; a useful audit also examines important statistics not directly used as model effects.
- Simulation checks expose structural mismatch without implying that a single p-value validates the complete model.

Degree-distribution goodness of fit

- The **out-degree distribution** counts how many ties actors send; for directed networks, the **in-degree distribution** counts how many ties actors receive.
- For undirected networks, the relevant quantity is the degree distribution.
- The app plots the observed frequency against the fitted-simulation mean and a simulation envelope.
- Systematic departures can reveal unmodeled popularity, activity heterogeneity, or a poor sparse-network baseline.

Triad-census goodness of fit

- A **directed triad census** counts the sixteen isomorphism classes of directed three-actor configurations.
- It checks patterns such as empty triads, mutual dyads, transitive triads, and cyclic triads beyond a single fitted closure statistic.
- A triad census is structural evidence, not a causal test for a social mechanism.
- The app provides the directed triad census for directed workflows; undirected workflows instead use compatible structural diagnostics such as degree and triangle/triad-related distributions.

All plots in the Session 3.1 app

- **Observed dynamics:** tie count and density by wave; successive-wave Jaccard overlap; maintained, formed, and dissolved ties; and direction-compatible reciprocity or closure summaries.
- **Estimation:** all coefficient estimates with approximate Wald intervals, a separate period-specific rate-parameter display, and convergence t -ratios for all effects.
- **Directed-network GOF:** out-degree, in-degree, directed triad census, reciprocity, geodesic distance, closure/two-path, and component/isolate diagnostics.
- **Undirected-network GOF:** degree, geodesic distance, shared-partner/triangle-closure, and component/isolate diagnostics; this is not a degree-only audit.
- **Distinct roles:** observed panels describe recorded change; intervals and convergence audit estimation; fitted simulations assess goodness of fit.

Worked public panel 1: Knecht classroom friendship

- **Data:** documented Dutch classroom friendship nominations at four observed waves.
- **Response:** directed binary nomination; the app retains actors observed at every selected wave.
- **Model:** rate parameters plus density, reciprocity, and directed transitive triplets.
- **Use:** inspect a compact multiwave panel before moving to the Day 3.2 behavior specifications built from the same release.
- **Limit:** classroom composition and missing-data rules must remain explicit; a coefficient is not a causal peer effect.

Worked public panel 2: RSiena s50 teaching excerpt

- **Data:** the public 50-actor, three-wave friendship teaching panel distributed with RSiena materials.
- **Response:** directed binary network with all 50 actors retained at the supplied waves.
- **Model:** the same transparent network-only SAOM is used to make cross-workflow differences visible.
- **Use:** a compact reproducible classroom calculation with simulation diagnostics and downloadable settings.
- **Limit:** the source is documented as a numerical teaching excerpt rather than a fully delineated population study.

Worked public panel 3: full Glasgow friendship

- **Data:** the public Teenage Friends and Lifestyle Study full Glasgow release with three waves.
- **Response:** friendship presence combines source codes 1=best friend and 2=just friend; code 10 is structural absence and is never recoded as zero.
- **Support:** Session 3.1 uses the documented 129-actor all-wave selection.
- **Use:** test how a larger repeated directed panel changes computation time, degree fit, and triad diagnostics.
- **Limit:** the model remains a conditional stochastic description of this documented cohort setting.

Worked public panel 4: Coleman high-school friendship

- **Data:** the public Coleman high-school friendship panel, observed in Fall 1957 and Spring 1958.
- **Response:** directed binary friendship nominations, restricted to actors observed at both endpoints.
- **Model:** one observed transition interval with density, reciprocity, and directed transitive-triplet effects.
- **Use:** contrast the information available from a two-wave panel with the richer multiwave examples.
- **Limit:** this release contains no verified repeated behavior suitable for Session 3.2 coevolution analysis.

Worked public panel 5: COW annual alliances

- **Data:** Correlates of War formal-alliance records assembled as annual binary treaty-presence snapshots for 1960–1970.
- **Response:** undirected alliance presence; the app restricts to states present at every selected annual endpoint.
- **Model:** undirected RSiena baseline with density and transitive-triads closure, not directed reciprocity.
- **Use:** make the directionality constraint operational rather than reusing a directed formula for an undirected network.
- **Limit:** annual snapshots do not reveal within-year ordering or treaty mechanism.

No-code workflow for a public panel

- ① Open the Render-hosted **Session 3.1** page and select a visible public workflow.
- ② Read the source, response definition, balanced-support rule, and interpretation limit.
- ③ Inspect observed tie counts and densities by wave before fitting.
- ④ Set the seed, n_3 , and number of goodness-of-fit simulations; retain the displayed settings in the record.
- ⑤ Run the RSiena calculation, then audit estimates, convergence ratios, and all available simulation envelopes.

Bring Your Own Data: required CSV structure

- **Nodes CSV:** columns `wave,id`; each actor occurs at most once per wave.
- **Edges CSV:** columns `wave,source,target`; the app rejects loops, duplicates, and endpoints absent at that wave.
- Declare whether the relation is **directed** or **undirected**; undirected dyads are canonicalized before validation.
- The app retains the intersection of actors observed in every selected wave and displays the resulting support.
- A valid upload is not evidence that a complex specification is identified; inspect waves, density, and convergence diagnostics.

Minimum reporting record for a network-only SAOM

- State the population, actor-support rule, tie response, directionality, wave dates, and missing-data treatment.
- List every rate and evaluation effect with its operational definition and the intended structural interpretation.
- Report estimation settings, seed, convergence summaries, warnings, and whether the run is a pilot or final analysis.
- Show simulation-based goodness-of-fit results for degree, compatible closure/triad structure, geodesic distance, and component/isolate summaries, including visible mismatches.
- Separate conditional model interpretation from causal language.

Limits of the Session 3.1 baseline

- A rate parameter does not identify the real temporal order of observed network changes.
- Density, reciprocity, and closure are deliberately limited structural terms; omitted heterogeneity or covariates can alter their fitted values.
- Balanced-panel restriction can change the target population and must be disclosed.
- Simulation-based fit checks diagnose discrepancies but do not establish model truth.
- Session 3.2 adds a repeated behavior and distinguishes network selection from behavioral influence under explicit additional assumptions.

Session 3.1 synthesis

- A SAOM treats repeated network observations as endpoints of a continuous-time actor-oriented process.
- The rate function selects opportunities; the evaluation function ranks local alternatives.
- Effects are interpreted through local change statistics and the stated choice set.
- Method-of-moments estimation requires simulation, convergence review, and simulation-based goodness-of-fit auditing.
- Responsible use begins with data support and ends with transparent limits, not with a coefficient table alone.

References



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